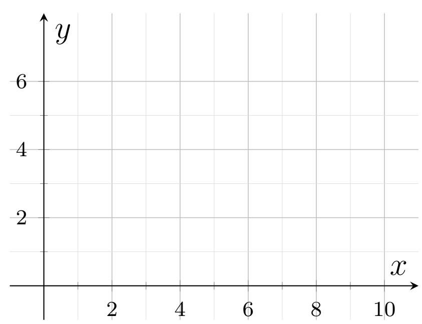
Perimeter and Area of Coordinate Polygons

Finding a polygon perimeter from coordinates, finding a polygon area from coordinates, and using slope or midpoint to justify a shape before measuring it

The three tools, and when each one earns its place.

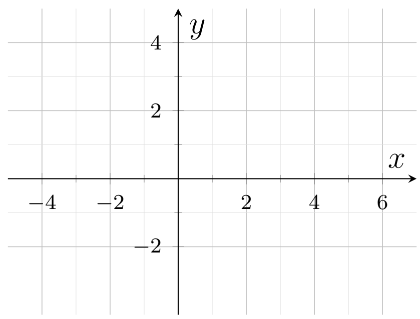
- **Distance** gives one side length: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. A perimeter is the sum of the side lengths, so find the ones you cannot read off the grid and add.
- **Shoelace** gives the area of any polygon straight from its vertices in order: $A = \frac{1}{2}\left|\sum(x_i y_{i+1} - x_{i+1} y_i)\right|$. Splitting the polygon into right triangles and rectangles gives the same number.
- **Slope and midpoint** do not measure anything. They justify a claim about the shape — perpendicular sides, bisected diagonals — which is often what makes a shortcut legal.
- Plot the polygon before computing. Every problem gives you a grid for it, and a sketch catches a mis-copied coordinate in seconds.
- Leave exact radicals unless the problem names a unit to round to, and label areas in square units.

1. A triangular sail is stitched along three straight seams. On the design grid its corners are $A(2, 1)$, $B(10, 1)$ and $C(10, 7)$, and one grid unit represents one metre. Plot the sail, find the length of the slanted seam AC, then find the total length of seaming around the sail.



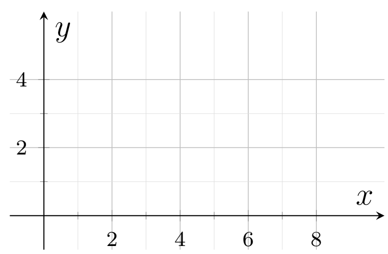
Answer: ________________ m

2. A rectangular concrete slab has corners at $(-3,-2)$, $(5,-2)$, $(5,3)$ and $(-3,3)$ on a site plan where one grid unit represents one metre. Plot the slab and find its area.



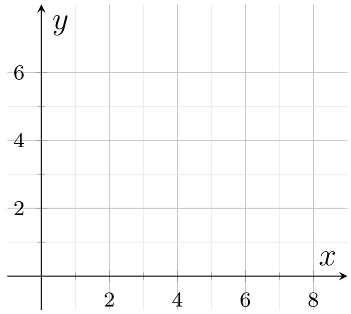
Answer: ________________ m^2

3. A parallelogram-shaped floor tile has vertices $P(0,0)$, $Q(6,0)$, $R(9,4)$ and $S(3,4)$, measured in grid units. Plot the tile, find the length of side QR, then find the perimeter of the tile.



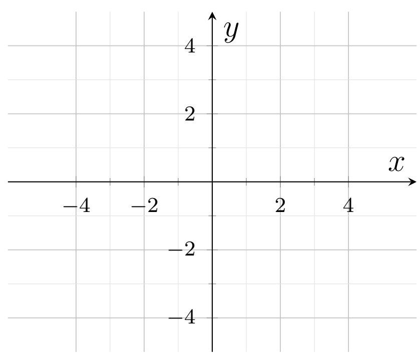
Answer: ________________ units

4. A triangular flowerbed has corners at $(0, 0)$, $(8, 2)$ and $(3, 7)$ in grid units. Plot the flowerbed and find its area.



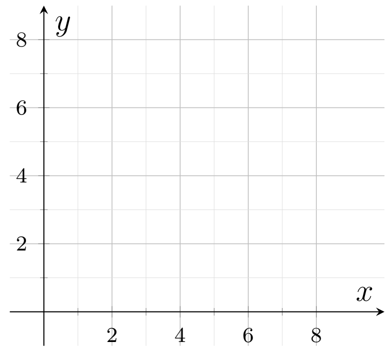
Answer: ________________ square units

5. A kite-shaped banner has corners $A(-4, 0)$, $B(0, 3)$, $C(4, 0)$ and $D(0, -3)$, in grid units. Plot it, then find the midpoint of diagonal AC and the midpoint of diagonal BD. Find the length of side AB. All four sides of this banner are the same length — use that to find its perimeter.



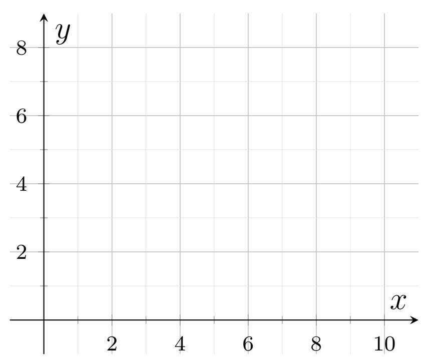
Answer: ________________ units

6. A triangular park has corners $L(1,2)$, $M(5,0)$ and $N(9,8)$, in grid units. Plot the park, then find the slope of LM and the slope of MN. Say in one sentence what those two slopes tell you about the corner at M, and use it to find the area of the park.



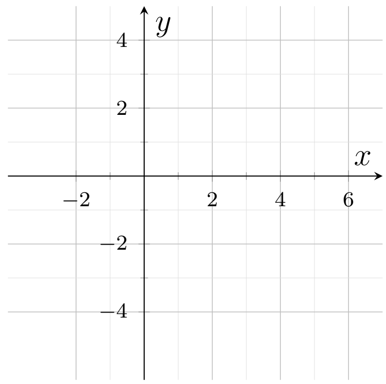
Answer: ________________ square units

7. A building lot has corners at $(0,0)$, $(10,0)$, $(10,6)$ and $(2,8)$ on a survey grid where one grid unit represents one metre. Plot the lot and find its area.



Answer: ________________ m^2

8. A designer claims that the quadrilateral with vertices $W(-2,1)$, $X(3,3)$, $Y(5,-2)$ and $Z(0,-4)$, in grid units, is a square. Plot it and find the length of side WX in exact radical form, then find the area of the quadrilateral. Congruent sides on their own would make the figure only a rhombus, so finish with a short written justification, using slopes, that decides whether the designer's claim is correct.



Answer: ________________ square units